MEDIA & ENTERTAINMENT INTELLIGENCE PROJECT

Netflix Content Strategy & Global Catalog Analytics

EXCEL | PYTHON (DATA CLEANING ONLY) | POWER BI

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1. Executive Summary

This project delivers a strategic catalog analysis of Netflix's global streaming titles (8,808 titles). Python (Pandas) was leveraged strictly for data auditing, null-value remediation, and dataset cleaning. The cleaned dataset was imported into Power BI to build interactive dashboard pages detailing content mix, age rating distribution, global production footprints, and multi-decade release trajectories.

TOTAL CATALOG TITLES	TOTAL MOVIES	TOTAL TV SHOWS
8,808	6,131 (69.62%)	2,676 (30.38%)

CLEANED MISSING VALUES	DOMINANT RATING	TOP CONTENT COUNTRY
4,307 Nulls	TV-MA	United States

Catalog Insight: Movies dominate the Netflix platform with **69.62%** of all titles versus **30.38%** for TV Shows. However, content strategy shifts in recent release years show accelerated expansion in serialized TV drama acquisitions across international markets.

2. Project Scope & Technology Stack

Business Context: Streaming platforms require accurate catalog categorization and demographic rating insights to guide licensing budgets, original content production, and localized market strategy.

TOOL / TECHNOLOGY	SPECIFIC PROJECT ROLE
Python (Pandas / NumPy)	Data ingestion, missing value detection (4,307 nulls), string imputation, and CSV cleaning export.
Power BI	Data modeling, DAX card metrics, interactive donut charts, genre distributions, rating funnels, and map visuals.
Excel / CSV	Supplementary data inspection and file validation downstream.

3. Data Cleaning & Preparation (Python Workflow)

An initial audit of `netflix_titles.csv` revealed 8,807 raw rows across 12 features with **4,307 total missing entries** concentrated across `director` (2,634), `cast` (825), `country` (831), `date_added` (10), `rating` (4), and `duration` (3).

```
import pandas as pd
import numpy as np

# Ingest raw Netflix titles dataset
data = pd.read_csv("netflix_titles.csv")

# Identify null distributions across attributes (4,307 total missing values)
null_counts = data.isnull().sum()

# Impute missing values with standardized categorical labels
data.fillna({
    "director": "Unknown",
    "cast": "Not Available",
    "country": "Unknown",
    "rating": "Unrated",
    "duration": "0",
    "date_added": "2000-01-01"
}, inplace=True)

# Export clean dataset for Power BI ingestion
data.to_csv("netflix_cleaned.csv", index=False)
```

Outcome: Clean dataset `netflix_cleaned.csv` exported with 0 null values, ready for Power BI visuals.

4. Power BI Dashboard: Catalog Overview & Ratings Breakdown

Page 1 of the Power BI dashboard suite visualizes the high-level split between Movies and TV Shows, genre volume distributions, and granular age rating metrics.

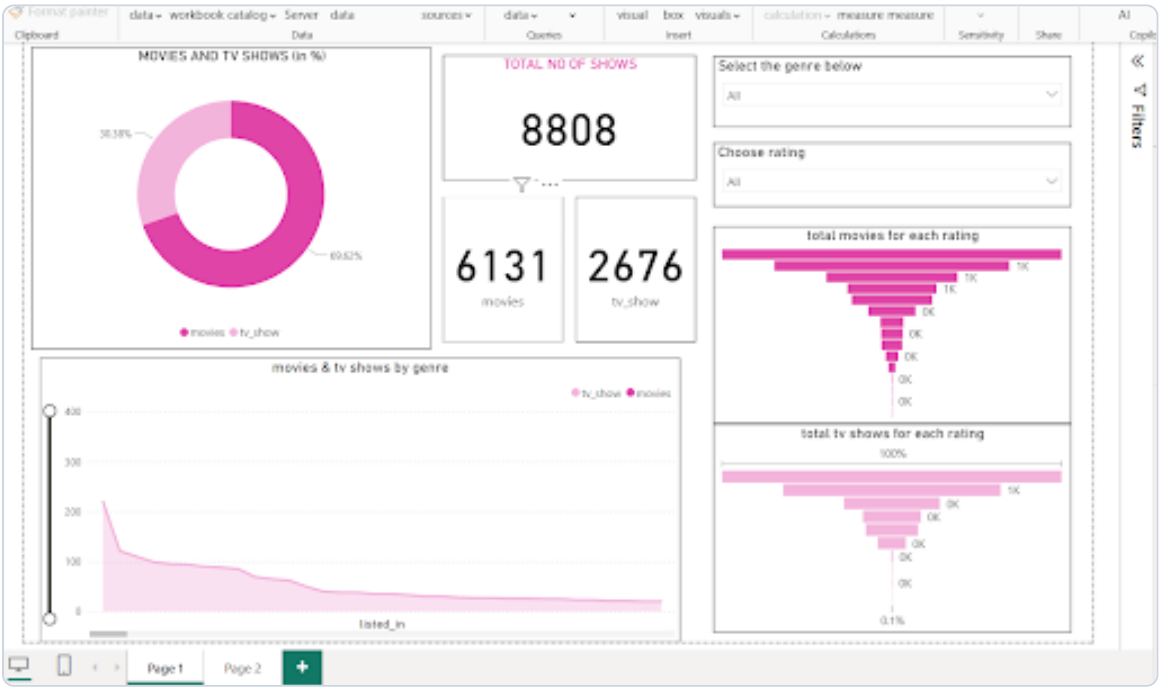


Figure 1: Power BI Dashboard Page 1 — Titles Split (Donut Chart), Total KPIs (8,808 Titles), Genre Area Trends, and Rating Funnels.

VISUAL TITLE	CHART TYPE	KEY DASHBOARD FINDING
Movies vs. TV Shows	Donut Chart	69.62% Movies (6,131 titles) vs. 30.38% TV Shows (2,676 titles).
Rating Distributions	Funnel / Bar Visuals	TV-MA is the largest rating tier for both Movies and TV Shows, followed by TV-14 and R.
Movies & TV Shows by Genre	Area Chart	International Movies, Dramas, and Comedies lead overall genre frequency.

5. Power BI Dashboard: Global Distribution & Release Trends

Page 2 explores geographic production output and historical content additions by release year.

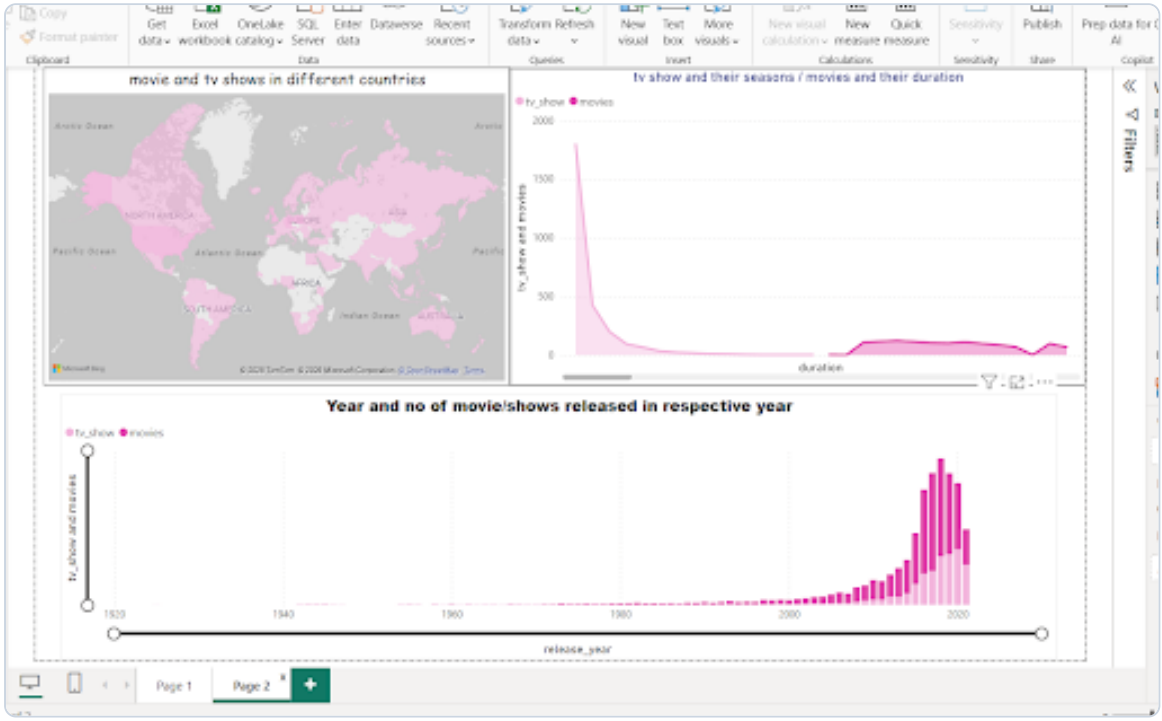


Figure 2: Power BI Dashboard Page 2 — Global Production Map, Season/Duration Trends, and Historical Release Timeline.

- **Geographic Concentration:** The United States and India represent the largest content source markets, followed by the UK, Japan, and South Korea.
- **Release Velocity:** Catalog growth accelerated sharply post-2015, peaking between 2018 and 2021 as Netflix expanded original productions globally.

6. Strategic Recommendations & Technical Skills Summary

DOMAIN	INSIGHT & ACTION PLAN
Catalog Strategy	Expand TV Show acquisitions to balance the 70/30 Movie ratio, capitalizing on higher multi-season subscriber retention rates.
Regional Investment	Increase local language originals across South-East Asia and LATAM, leveraging established viewership trends in India and East Asia.
Technical Stack	Demonstrates end-to-end analytics combining **Python data cleaning** with **Power BI interactive reporting**.